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E-430 SURGE ARRESTERS

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing of Surge Arrester in accordance with Part 4, Subsection 1.2 Summary of Work.

2. CODES AND STANDARDS

The Contactor shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards” and the following specific standards:

2.1 Indonesian National Standards

PUIL : Peraturan Umum Instalasi Listrik-Indonesia.

SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

IEC 60060 : High-voltage test techniques

IEC 60099 : Surge arresters

IEC 62155 : Hollow pressurized and unpressurized ceramic and glass insulators for use in electrical equipment with rated voltages greater than 1000 V

2.3 National Electrical Manufacturers Association

NEMA 107 : Methods of Measurements of Radio Influence Voltage (RIV) of High Voltage Apparatus

2.4 Institute of Electrical and Electronics Engineers

ANSI/IEEE C62.11 : Metal-Oxide Surge Arrester for AC Power Circuits

3. OPERATING AND DESIGN CONDITIONS

Surge Arresters shall be designed for outdoor operation located in a tropical climate along the sea coast. Refer to Part 4, Section 1.1, 2.14 and 2.16.

4. DETAILED REQUIREMENTS

4.1 General

Each arrester unit shall have a pressure relief device for relieving internal gas pressure prior to porcelain rupture in the event to arrester failure.

The surge arrester shall be in general accord with IEC 60099.

Surge arresters shall be gapless metal oxide type.

4.2 Ratings

- (1) Arrester rating (RMS rated voltage)..... 444 kV
- (2) System Ratings:
 - a. Nominal voltage level..... 500 kV
 - b. Maximum voltage level..... 550 kV
 - c. Frequency..... 50 Hz
- (3) Maximum switching impulse protective level..... 900 kV
- (4) Maximum residual voltage at discharge current (wave 8/20 μ s)
 - a. At 10,000 A..... 977 kV
 - b. At 20,000 A..... 1,060 kV
- (5) Duty cycle current 20 kA
- (6) High current short duration (minimum crest) 100 kA
- (7) Long duration discharge class..... 5
(as per IEC 60099 or equivalent)
- (8) Phase-to-ground insulation:
 - a. Impulse withstand voltage (1.2 x 50 μ s) crest 1,550 kV
 - b. 50 Hz dry (and wet*) withstand test voltage..... 680 kV
 - c. Switching impulse withstand 1,175 kV
- (9) Radio Interference Voltage:

The maximum RIV at 1 MHz shall be:

 - a. Applied phase-to-ground voltage 333 kV
 - b. RIV 2,500 μ volts

The measurements shall be in accordance with IEC 60099-4. In making the RIV test, radio noise readings shall be taken at various points so as to determine the break points.

4.3 Surge Counter and Leakage Current Meter

A surge counter and leakage current meter shall be furnished with each arrester and shall operate for each surge current pulse or leakage current, respectively through the arresters and shall be designed for continuous service.

The surge counter and leakage current meter shall be designed for outdoor service or a weatherproof housing with window for reading the counter. Interconnecting leads between the arrester, surge counter, current leakage meter and station ground shall be provided.

The surge counter shall be connected with bolted connections such that a link can be connected across the surge counter to ground the surge arrester prior to removing surge counter for maintenance or adjustment. The surge counter shall be designed to prevent personnel injury during the arrester operation.

4.4 Corona

Surfaces in contacts with live potential shall contain no projection points or irregularities and they shall be designed for a minimum of visible corona or brush discharge when the arrester is completely assembled.

No visual corona shall exist at applied voltage. Lights shall be turned off in the tests bay, and visual start and stop corona voltages shall be recorded with the corresponding radio noise voltages on both ascending and descending voltage values. Photographs shall be taken for each recorded voltage value. NEMA 107 test procedures shall be used.

4.5 Porcelain

Contractor shall meet the requirements of IEC 62155 for all porcelain used in these surge arresters. Porcelain insulation for surge arresters shall be chocolate brown color and shall be permanently marked.

4.6 Nameplates

Nameplates shall be as per IEC 60099. In addition to the overall nameplate, a nameplate for each individual unit shall be integrally fixed to this unit.

5. PERFORMANCE TESTING AND GUARANTEES

The surge arresters shall be tested in accordance with IEC 60099-4 or equivalent.

The temporary over voltage (TOV) capability curve for the surge arrester shall be submitted. The curve shall include TOV rating with no prior absorption and with maximum energy absorption.

6. QUALITY ASSURANCE/QUALITY CONTROL

The quality assurance/quality control for surge arrester shall conform to the requirements included in Part 4, Section 2, Subsection 2.7.

The quality assurance/quality control procedures specified shall included, but not be limited to, the followings:

- (1) All routine tests specified in Clause 5 of this specification
- (2) All special type tests performed for the specification as per Clause 5.

The quality assurance/quality control documentation as specified shall included, but not be limited to, the followings:

- (1) Shop Test Report.
- (2) Al other documentation required by applicable codes and standards.

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